ENSEMBLE MACHINE LEARNING APPROACHES FOR HEART DISEASE RISK PREDICTION USING CLINICAL DATA

7COM1039 – MSc Project

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1. Background Research and Literature Review

Heart disease (cardiovascular disease, CVD) remains a leading global cause of mortality and creates heavy pressure on healthcare systems. A key challenge is that diagnosis requires interpreting multiple clinical and demographic factors (e.g., age, blood pressure, cholesterol, chest pain characteristics) that may interact in non-linear ways. Machine learning (ML) can learn these relationships from data and support earlier risk identification. However, a single ML model can be unstable (high variance), biased (high bias), or overly sensitive to noise and sampling. Ensemble learning aims to reduce these weaknesses by combining multiple models to obtain more robust and reliable predictions (Dietterich, 2000).

1.1 Aim

The aim of this project is to develop and evaluate ensemble machine learning models for predicting heart disease risk using clinical and demographic patient data. The project will compare ensemble approaches against individual machine learning models, focusing on predictive accuracy, robustness, and suitability for healthcare decision-support.

1.2 Research Questions

RQ1: What are the best design choices for building a bagging-based ensemble classifier for heart disease risk prediction (i.e., choice of base learners, how diversity is created via sampling/parameters/features, and how outputs are combined)?

RQ2: What trade-offs exist between predictive accuracy and model interpretability when using ensemble methods in healthcare, and how can explainability techniques mitigate these trade-offs?

1.3 Objectives

- Review relevant literature on heart disease prediction and ensemble machine learning methods.
- Acquire and preprocess a publicly available heart disease dataset.
- Implement baseline (single) models and bagging/ensemble variants.
- Evaluate models using appropriate metrics (Accuracy, Precision, Recall, F1-score, ROC-AUC) and cross-validation.
- Analyse results, including robustness and interpretability, and discuss healthcare implications.

1.4 Literature Search Strategy

A structured literature search was conducted using IEEE Xplore. Search strings included combinations of: “heart disease prediction”, “cardiovascular”, “ensemble”, “bagging”, “voting”, “stacking”, “feature selection”, and “clinical data”. The year range was set to 2021–2024 to prioritise recent advances and evaluation practices. Titles and abstracts were screened first, followed by full-text review of the most relevant papers.

1.5 Inclusion and Exclusion Criteria

Inclusion Criteria	Exclusion Criteria
• Published 2021–2024 • Peer-reviewed journal or IEEE conference • Focuses on heart disease / cardiovascular risk prediction • Uses structured clinical/demographic data (tabular) • Reports quantitative evaluation metrics	• Published before 2020 • Non-peer-reviewed sources (blogs, forums) • Not focused on cardiovascular outcomes • Pure signal-only work without clinical variables (unless directly relevant) • No evaluation metrics / no experimental evidence

1.6 Critical Literature Review

The reviewed literature consistently shows that ensemble approaches can outperform single classifiers on heart disease datasets, but results vary depending on dataset size, class imbalance, feature engineering, and validation design. Therefore, my project focuses on the "design choices" behind ensembles (how base learners are built and how their predictions are combined), rather than only reporting a final accuracy number.

Paper A: Feature Selection + Ensemble Learning for Heart Disease (2021)

Lakshmanarao et al. (2021) proposed a pipeline combining preprocessing, feature selection (ANOVA and mutual information), sampling methods for imbalance (ROS, SMOTE, ADASYN), and several classifiers including ensemble voting and stacking. They reported strong performance, including high accuracy for stacking on an imbalanced dataset after oversampling. The strength of this paper is its practical focus on dataset imbalance and feature selection, which are realistic issues in clinical data. A limitation is that extremely high scores after oversampling can sometimes reflect leakage or overfitting if validation is not carefully stratified. This paper directly informs my planned experiments: (i) compare feature selection strategies, and (ii) evaluate imbalance handling before building ensembles.

Paper B: Random Forest with Sequential Feature Selection (2024)

Gupta et al. (2024) compared multiple ML methods and found Random Forest to perform best, reporting very high accuracy when Sequential Feature Selection was applied. The paper reinforces that feature selection can simplify the model and can increase performance; however, results near 100% should be treated cautiously because small datasets may inflate accuracy. For my project, the takeaway is to validate using stratified k-fold cross-validation and to report multiple metrics beyond accuracy (e.g., ROC-AUC and recall) to avoid misleading conclusions.

Paper C: Supervised ML Prediction on UCI Cleveland (2021)

Hossen et al. (2021) evaluated Logistic Regression, Decision Tree, and Random Forest on the UCI Cleveland dataset and reported strong performance for Logistic Regression. This is important because it shows simpler and more interpretable models can be competitive on small, structured datasets. It also motivates my RQ2: if a simpler model performs similarly to an ensemble, interpretability may justify choosing the simpler model in healthcare contexts.

Paper D: Naive Bayes System and Imbalance Effects (2023)

Sabri et al. (2023) presented a Naive Bayes-based web system, highlighting how class imbalance can distort accuracy. They noted that balancing reduced headline accuracy but improved fairness across classes. This supports my plan to focus on recall, F1-score and ROC-AUC, and to include confusion matrices to understand false negatives (high-risk patients predicted as low-risk), which is the more harmful error in medical screening.

1.7 Practical Research and Technical Resolution Examples

The literature also guided practical design decisions. For example, multiple papers emphasised that clinical datasets can be imbalanced, motivating evaluation using stratified splits and resampling (Lakshmanarao et al., 2021; Sabri et al., 2023). Similarly, feature selection was repeatedly used to reduce dimensionality and improve performance (Lakshmanarao et al., 2021; Gupta et al., 2024). As a result, my implementation prioritises: (i) reproducible preprocessing, (ii) stratified k-fold validation, and (iii) an ensemble design that explicitly creates diversity across base learners via sampling and parameter variation.

2. Summary of Progress to Date

Progress to date has focused on establishing a reliable experimental pipeline and producing evidence of early implementation. The work completed aligns directly with the project objectives: literature review, dataset acquisition and preprocessing, baseline model implementation, and initial evaluation setup.

2.1 Work Completed

- Completed structured literature search on IEEE Xplore and selected core papers aligned with ensemble design and evaluation.
- Selected a publicly available heart disease dataset (UCI Cleveland) and documented its features and target definition.
- Implemented preprocessing steps (encoding, scaling, handling missing values where applicable) and reproducible train/test splits.
- Implemented baseline models: Logistic Regression, Decision Tree, Naive Bayes, Support Vector Machine, and Random Forest.
- Configured evaluation pipeline using stratified k-fold cross-validation, confusion matrix, and ROC-AUC reporting.

2.2 Tools, Technologies and Justification

Python was selected as the implementation language due to its maturity in data science and availability of validated libraries. scikit-learn supports consistent model training, cross-validation, and metric reporting, reducing the risk of implementation errors. pandas and NumPy provide reliable data manipulation. GitHub version control is used to track progress and ensure reproducibility.

2.3 Evidence of Progress (to be included in Appendices)

Appendix 1: Dataset description and feature dictionary (UCI Cleveland).
Appendix 2: Preprocessing pipeline summary (encoding, scaling, split strategy).
Appendix 3: Baseline model training scripts and outputs (accuracy, ROC-AUC).
Appendix 4: Confusion matrices and classification reports for baseline models.
Appendix 5: Git commit log screenshots / repository activity summary.

2.4 Challenges and How They Were Handled

The main challenges encountered so far are (i) small dataset size and potential overfitting, and (ii) selecting evaluation metrics that reflect clinical risk. These have been handled by adopting stratified k-fold cross-validation and reporting multiple metrics. In addition, the project will treat recall for the positive class as a key indicator because false negatives are more harmful in screening-style applications.

3. Ethical, Legal, Professional and Social Issues

This project uses secondary, publicly available clinical data and does not involve direct contact with patients or collection of identifiable personal data. Therefore, formal ethics approval is not expected to be required. The justification is that the dataset is already anonymised and is used strictly for academic research and model evaluation, not for clinical deployment.

3.1 Ethical Issues

Ethically, the main risks relate to harm from incorrect predictions and unfair performance across subgroups. A false negative (high-risk predicted as low-risk) could delay medical attention. To address this, evaluation will prioritise recall and will include confusion matrices and ROC-AUC rather than accuracy alone. Fairness considerations will be explored by checking performance differences across available demographic variables, where the dataset permits.

3.2 Legal Issues

Legally, the project must comply with data protection expectations such as the UK GDPR where applicable. As the dataset is public and anonymised, the main legal duty is to ensure secure handling (no re-identification attempts) and correct attribution of data sources. Intellectual property is respected through correct citation of external code, datasets and academic sources.

3.3 Professional Issues

Professionally, the project will follow reproducible research practice: clear documentation, version control, consistent experiments, and transparent reporting of limitations. This includes reporting negative results or cases where ensembles do not outperform simpler models, because reliable conclusions are more important than impressive numbers.

3.4 Social Issues

Socially, decision-support tools can improve access to early screening, but they can also increase inequality if models perform worse for underrepresented groups. Therefore, interpretability and transparent performance reporting are important. Explainability techniques such as SHAP and LIME can help clinicians understand why the model predicts risk, supporting trust and responsible use (Sathyan et al., 2022).

4. Project Plan

A hybrid project management approach is adopted, combining a structured timeline with iterative experimentation. Weekly milestones and supervisor feedback will be used to control scope and ensure progress. The plan reserves dedicated time for testing, validation, and final report writing to meet MSc standards.

4.1 Remaining Tasks and Target Dates

Task	Deliverable	Target Date
Implement bagging ensembles (bootstrap sampling + base learners)	Bagging experiment scripts + results table	[Week/Date]
Design diversity strategies (sampling / feature subsampling / parameter variation)	Diversity ablation study results	[Week/Date]
Implement aggregation methods (majority voting, weighted voting)	Comparison of combination rules	[Week/Date]
Hyperparameter optimisation (GridSearchCV/RandomSearchCV)	Best model configs + justification	[Week/Date]
Interpretability study (baseline vs ensemble + SHAP/LIME)	Explainability examples + discussion	[Week/Date]
Final evaluation and robustness checks (k-fold, ROC-AUC, calibration)	Final performance and robustness report	[Week/Date]
Write final dissertation + prepare demo	Complete dissertation and presentation/demo	[Week/Date]

4.2 Risk, Quality and Feedback Strategy

Key risks include overfitting due to limited dataset size, and misleading conclusions from relying on a single metric. To mitigate this, the project will use stratified cross-validation, multiple metrics, and calibration checks. Quality will be assessed through reproducibility (fixed random seeds and saved configurations), comparative baselines, and clear reporting. Supervisor meetings will be used to review progress, adjust priorities, and confirm that experiments remain aligned with the research questions.

5. Level of the Project (MSc Justification)

This project meets MSc level expectations by combining theory-driven investigation with rigorous experimentation and validation. The work goes beyond applying a single classifier: it systematically studies ensemble design choices (base learner selection, diversity creation, combination rules) and evaluates their impact on robustness and reliability. It also addresses a key healthcare requirement: balancing predictive performance with interpretability and accountability. The planned evaluation includes ablation studies, cross-validation, and explainability analyses, which demonstrate depth, complexity, and technical rigour expected at MSc level.

References (Harvard)

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Appendices

Appendix 1: Dataset source and feature dictionary (UCI Cleveland)

[Insert evidence here]

Appendix 2: Preprocessing pipeline screenshots / logs

[Insert evidence here]

Appendix 3: Baseline model code snippets and outputs

[Insert evidence here]

Appendix 4: Confusion matrices, ROC curves, and metric tables

[Insert evidence here]

Appendix 5: Git repository activity log / commit history

[Insert evidence here]

Appendix 6: Supervisor meeting notes and action points

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